

BUSINESS REQUIREMENT DOCUMENT (BRD)

Help Desk Management System

**Help Desk
Management
Services**



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Tables of content

1. Introduction.....	2
1.1 Background.....	2
1.2 Project Objectives.....	2
2. Scope of Work.....	2
2.1 Functional Scope.....	2
2.2 System Users.....	2
3. Functional Requirements.....	3
3.1 ICT Register.....	3
3.2 Server Room Access Register.....	3
3.3 ICT-End User Logs.....	3
3.4 ICT System Monitoring.....	3
3.5 Laptop/Computer Register & Asset Movement.....	3
3.6 ICT System Documentation Management.....	4
4. Non-Functional Requirements.....	4
5. Use Cases.....	4
6. System Architecture.....	5
7. Acceptance Criteria.....	5
8. Glossary.....	5

1. Introduction

1.1 Background

The National Crime Research Centre requires an **automated Help Desk Management System** to enhance efficiency, streamline records management, and eliminate manual documentation processes in the ICT department. This system will digitize the existing ICT logs, asset tracking, monitoring, and end-user support while ensuring security and compliance.

1.2 Project Objectives

- **Automate manual record-keeping** for ICT operations.
- **Improve asset tracking** through QR code and digital registers.
- **Enhance security** via role-based access control.
- **Monitor ICT systems** in real time and generate reports.
- **Improve issue tracking** through a ticketing system for user requests.

2. Scope of Work

2.1 Functional Scope

The system will support:

- **Ticketing and issue tracking** for ICT support requests.
- **Digital record-keeping** for ICT logs, including asset movement and access logs.
- **Automated system monitoring** with real-time alerts.
- **User and role management** for controlled access to ICT resources.
- **Audit and reporting** functionalities.

2.2 System Users

- ICT Staff
- End Users (Employees requesting support)
- Security Personnel (for access logs)
- System Administrators

3. Functional Requirements

3.1 ICT Register (Help Desk Ticketing System)

- Users can create tickets for ICT issues.
- Automated assignment of tickets to IT staff.
- Ticket status tracking and resolution logs.
- Email/SMS notifications for updates.

3.2 Server Room Access Register

- Digital logs of entry/exit with timestamps.
- Alerts for unauthorized access attempts.

3.3 ICT-End User Logs

- Digital record of user-reported issues and resolutions.
- AI chatbot for automated troubleshooting of common issues.
- Historical logs for analysis and performance tracking.

3.4 ICT System Monitoring

- Real-time monitoring of servers, networks, and databases.
- Alerts for security breaches and system failures.
- Automated reports on performance trends.

3.5 Laptop/computer register and ICT Asset Movement

- QR code tagging for assets.
- Digital check-in/check-out logs.
- Maintenance history tracking.

3.6 ICT System Documentation Management

- Centralized repository for IT policies, guides, and documentation.
- Role-based access to sensitive documents.
- Search/filter functionality for quick document retrieval.

4. Non-Functional Requirements

- **Security:** Data encryption, access control, and audit logging.
- **Performance:** System should support 1,000+ users with minimal latency.
- **Usability:** Web-based and mobile-friendly interface.
- **Scalability:** Cloud-compatible architecture for future expansion.
- **Compliance:** Adherence to data protection regulations.

5. Use Cases

Use Case 1: ICT Help Desk Ticketing

- **Actors:** Employee, IT Staff
- **Description:** An employee submits a support request, which gets assigned and resolved.
- **Precondition:** The employee is logged into the system.
- **Post-condition:** The issue is resolved, and the ticket is closed.

Use Case 2: Server Room Access Log

- **Actors:** Security Personnel, IT Staff
- **Description:** Logs staff entries into the server room and alerts unauthorized attempts.
- **Precondition:** The staff member has valid access credentials.

- Post-condition: Access log is recorded securely.

6. System Architecture

- **Frontend:** React / Next.js for UI.
- **Backend:** Java / Spring boot for API handling.
- **Database:** PostgreSQL / MySQL for structured data storage.
- **Security:** Access control through OAuth authentication, backing up data and using 2FA or Multifactor Authentication.
- **Hosting:** Cloud-based deployment on AWS/Azure.

7. Acceptance Criteria

- Successful automation of all six manual logs.
- 99.9% uptime with real-time system monitoring.
- Secure handling of user data and system logs.
- End-user satisfaction rate of 90% in usability tests.

8. Glossary

- **OAuth:** Open authorization standard for secure logins.
- **QR Code:** Machine-readable code used for asset identification.
- **AI Chatbot:** Automated assistant for troubleshooting ICT issues.